OurMocks

2027 CYCLE

MOCK · PRACTICE · CRACK

CUET UG

Chemistry

Alcohols, Phenols and Ethers

Official Syllabus Unit VII · Class 12 · CUET UG 2027 Cycle Notes

INSIDE
Concept Notes

INSIDE
Solved Examples

INSIDE
Practice MCQs + Solutions

Aligned to the latest official syllabus · Concept-first explanations

www.ourmocks.com

Syllabus Alignment

Sub-topic	Official syllabus pointer (NTA CUET(UG) Chemistry syllabus, the current 2027 baseline, Unit VII)	Skills the questions test
Classification & Structures	Classification of Alcohols, Phenols and Ethers (mono-, di-, tri- or polyhydric; 1°/2°/3°), IUPAC nomenclature, electronic structures of functional groups ($C-O-H$ and $C-O-C$ bond angles).	Classifying hydroxy/alkoxy compounds, assigning IUPAC names, explaining why the $C-O-C$ angle in ethers exceeds the tetrahedral angle (111.7°).
Preparation of Alcohols & Phenols	Methods of preparation of Alcohols (hydration of alkenes, hydroboration-oxidation, reduction of carbonyls and esters, Grignard additions) and Phenols (from haloarenes, benzenesulphonic acid, diazonium salts, and cumene).	Applying Markovnikov vs anti-Markovnikov syn addition, distinguishing $NaBH_4$ vs $LiAlH_4$ selectivity, and detailing the industrial cumene-to-phenol process.
Physical & Chemical Properties	Physical properties (boiling points, intermolecular hydrogen bonding, solubility). Chemical reactions of alcohols (acidity, esterification, reactions involving $C-O$ cleavage, Lucas test, Victor Meyer test, dehydrogenation over Cu, dehydration).	Ordering boiling points and solubilities, predicting Lucas turbidity times, calculating esterification rates, and identifying dehydration alkene products.
Chemistry of Phenols	Reactivity of Phenols: acidity and substituent effects ($-NO_2$ vs $-CH_3$), electrophilic aromatic substitutions (halogenation, nitration), Kolbe's reaction, Reimer-Tiemann reaction, reaction with zinc dust, oxidation.	Explaining phenoxide resonance stability, separating o- and p-nitrophenol by steam distillation, deducing dichlorocarbene intermediate, and synthesizing aspirin.
Ethers: Synthesis & Cleavage	Methods of preparation of Ethers (acid-catalysed dehydration of alcohols, Williamson ether synthesis), physical properties, chemical reactions (cleavage of $C-O$ bond by hydrogen halides, electrophilic substitution of anisole).	Executing Williamson syntheses with primary halides vs avoiding 3° elimination, predicting HI cleavage products (S_N2 vs S_N1), and directing anisole substitutions.

Sub-topic	Official syllabus pointer (NTA CUET(UG) Chemistry syllabus, the current 2027 baseline, Unit VII)	Skills the questions test
Commercial Alcohols	Manufacture, uses, and physiological hazards of Methanol (CH_3 OH) and Ethanol (C_2H_5 OH), denatured alcohol.	Detailing water gas catalytic synthesis of methanol, yeast enzyme fermentation steps (invertase and zymase), and physiological toxicity mechanisms.

EXAM PATTERN

CUET UG Paper Pattern & Weightage: The Chemistry paper carries **50 compulsory questions** in **60 minutes** (+5 marks for correct, −1 mark for incorrect). Unit VII (Alcohols, Phenols and Ethers) contributes 4 to 6 questions focusing on Lucas & Victor Meyer distinctions, Williamson synthesis with 3° halides, Kolbe and Reimer-Tiemann mechanisms, ether cleavage with HI, and phenol substituent acidity trends.

Module 1: Classification, Nomenclature & Structural Geometry

1. Classification & Nomenclature

1. **Alcohols:**
   - **By Hydroxyl Group Count:** Monohydric (ethanol), Dihydric / Glycols (ethane-1,2-diol), Trihydric / Glycerols (propane-1,2,3-triol), and Polyhydric (sorbitol, mannitol).
   - **By Carbon Hybridisation ($C(sp^3) - OH$):**
     - **Primary (1°):** −OH attached to $-CH_2-$ (e.g. ethanol, propan-1-ol).
     - **Secondary (2°):** −OH attached to $>CH-$ (e.g. propan-2-ol, butan-2-ol).
     - **Tertiary (3°):** −OH attached to $>C-$ (e.g. 2-methylpropan-2-ol).
     - **Allylic Alcohols:** −OH bonded to sp^3 carbon adjacent to $C=C$ ($CH_2=CH-CH_2$ OH).
     - **Benzylic Alcohols:** −OH bonded to sp^3 carbon adjacent to benzene ring ($C_6H_5CH_2$ OH).
   - **Vinylic Alcohols ($C(sp^2) - OH$):** −OH attached directly to $C=C$ (e.g. vinyl alcohol, $CH_2=CH$-OH, which rapidly tautomerises to acetaldehyde).
2. **Phenols:** Hydroxyl group is bonded directly to an aromatic benzene ring (Ar-OH).
   - Monohydric: Phenol (C_6H_5 OH), o-cresol (2-methylphenol), m-cresol, p-cresol.
   - Dihydric: Catechol (benzene-1,2-diol), Resorcinol (benzene-1,3-diol), Quinol / Hydroquinone (benzene-1,4-diol).
   - Trihydric: Pyrogallol (benzene-1,2,3-triol), Hydroxyquinol (benzene-1,2,4-triol), Phloroglucinol (benzene-1,3,5-triol).
3. **Ethers:** Compounds containing an alkoxy or aryloxy group ($R-O-R'$ or $R-O-$Ar).
   - **Symmetrical / Simple Ethers:** Both alkyl groups are identical (e.g. diethyl ether, $C_2H_5-O-C_2H_5$).

- **Unsymmetrical / Mixed Ethers:** Alkyl/aryl groups differ (e.g. ethyl methyl ether, $C_2H_5 - O - CH_3$; anisole, $C_6H_5 - O - CH_3$).

2. Molecular Geometry & Bond Angle Comparison

In alcohols, phenols, and ethers, the oxygen atom is sp^3-hybridised with two bonding pairs and two non-bonding lone pairs:

- **Methanol ($C - O - H$):** The bond angle is 108.9°, slightly smaller than the ideal tetrahedral angle (109.5°). This slight compression is caused by the greater electrostatic repulsion exerted by the two unshared lone pairs on oxygen against the bonding pairs (**lone pair - bonding pair repulsion**).
- **Phenol ($C - O - H$):** The bond angle is 109°. The $C - O$ bond length in phenol (136 pm) is substantially shorter than in methanol (142 pm). This shortening arises from:
  1. **Resonance Effect:** Partial double-bond character between ring carbon and oxygen.
  2. **Hybridisation Difference:** Carbon is sp^2-hybridised (33% s-character), holding electrons tighter.
- **Dimethyl Ether ($C - O - C$):** The bond angle is 111.7°, significantly **larger than the tetrahedral angle** (109.5°). **Critical Examination Fact:** The repulsive steric interaction between the two bulky, electron-rich alkyl groups outweighs the lone-pair compression, forcing the $C - O - C$ bond angle to widen to 111.7°.

Module 2: Preparation of Alcohols and Phenols

1. Synthesis of Alcohols

1. **From Alkenes:**
   - **Acid-Catalysed Hydration:** Direct addition of water in the presence of dilute mineral acid (H_2SO_4):

     $$CH_3 - CH = CH_2 + H_2O \xrightarrow{H^+} CH_3 - CH(OH) - CH_3 \quad \text{(Markovnikov addition)}$$

     Mechanism: Stepwise addition via a planar carbocation intermediate. Prone to 1,2-hydride or 1,2-methyl shifts to yield rearranged alcohols.
   - **Hydroboration-Oxidation (H.C. Brown, Nobel Prize 1979):** Diborane (B_2H_6) reacts with alkenes to form trialkylboranes, which are oxidized by alkaline hydrogen peroxide ($H_2\frac{O_2}{OH^-}$):

     $$3CH_3 - CH = CH_2 + (BH_3)_2 \rightarrow (CH_3CH_2CH_2)_3B \xrightarrow{H_2\frac{O_2}{OH^-}} 3CH_3CH_2CH_2\,OH + B(OH)_3$$

     CUET Hallmark Rule: Overall results in **anti-Markovnikov addition of water** via concerted **syn-addition**. Because no carbocation is formed, **no structural rearrangements occur**.
2. **From Carbonyl Compounds:**
   - **Reduction of Aldehydes and Ketones:**
     - Aldehydes are reduced to primary (1°) alcohols; ketones are reduced to secondary (2°) alcohols.
     - Reagents: Sodium borohydride ($NaBH_4$) in ethanol or Lithium aluminium hydride ($LiAlH_4$) in dry ether.
   - **Reduction of Carboxylic Acids & Esters:**

- Carboxylic acids are reduced exclusively to 1° alcohols by the powerful reducing agent $LiAlH_4$.
- Commercially, acids are converted to esters and reduced via catalytic hydrogenation (**Bouveault-Blanc Reduction**):

$$R-\text{COOH} \xrightarrow{\text{R'OH / H}^+} R-\text{COOR'} \xrightarrow[\text{Catalyst}]{\text{H}_2} R-\text{CH}_2\ \text{OH} + R'-\text{OH}$$

3. **From Grignard Reagents (Carbon-Carbon Chain Extension):** Nucleophilic addition of $R-\text{MgX}$ to carbonyl groups followed by acidic hydrolysis:
   - **Methanal (Formaldehyde, HCHO):** Yields **Primary (1°) Alcohols**:

$$\text{HCHO} + R-\text{MgX} \rightarrow R-\text{CH}_2\ \text{OMgX} \xrightarrow{\text{H}_2\text{O}} R-\text{CH}_2\ \text{OH} + \text{Mg(OH)}X$$

   - **Other Aldehydes ($R'-$CHO):** Yields **Secondary (2°) Alcohols**:

$$R'-\text{CHO} + R-\text{MgX} \rightarrow R'-\text{CH(OMgX)}-R \xrightarrow{\text{H}_2\text{O}} R'-\text{CH(OH)}-R$$

   - **Ketones (R'_2 C=O):** Yields **Tertiary (3°) Alcohols**:

$$R'_2\ \text{C=O} + R-\text{MgX} \rightarrow R'_2C(\text{OMgX})-R \xrightarrow{\text{H}_2\text{O}} R'_2C(\text{OH})-R$$

2. Industrial & Laboratory Synthesis of Phenols

1. **From Cumene (Isopropylbenzene — Dominant Commercial Route):** Over 90% of global synthetic phenol is manufactured via this single elegant process:
   - Cumene is oxidized by passing air or oxygen through it at $368-408$ K to produce **cumene hydroperoxide**.
   - Cumene hydroperoxide is treated with dilute sulphuric acid (H_2SO_4) at $323-363$ K:

$$C_6H_5-\text{CH(CH}_3)_2 + O_2 \rightarrow C_6H_5-C(\text{CH}_3)_2-\text{O-O-H} \xrightarrow{\text{H}^+} C_6H_5\ \text{OH} + \text{CH}_3\text{COCH}_3$$

   Industrial Merit: Generates equimolar quantities of **Acetone (propan-2-one)** as a high-value commercial byproduct.
2. **From Haloarenes (Dow's Process):** Chlorobenzene is heated with $6-8\%$ aqueous sodium hydroxide at 623 K under 300 atm pressure, followed by acidification:

$$C_6H_5\ \text{Cl} + 2\ \text{NaOH} \xrightarrow{623\ \text{K},300\ \text{atm}} C_6H_5\ \text{ONa} + \text{NaCl} + \text{H}_2\text{O} \xrightarrow{\text{HCl}} C_6H_5\ \text{OH}$$

3. **From Benzenediazonium Salts:** Benzenediazonium chloride is warmed gently with water or dilute mineral acid:

$$C_6H_5-N_2^+Cl^- + \text{H}_2\text{O} \xrightarrow{\text{warm}} C_6H_5\ \text{OH} + N_2\uparrow + \text{HCl}$$

4. **From Benzenesulphonic Acid:** Benzene is sulphonated with oleum; sodium benzenesulphonate is fused with solid sodium hydroxide at 573 K, yielding sodium phenoxide which on acidification gives phenol.

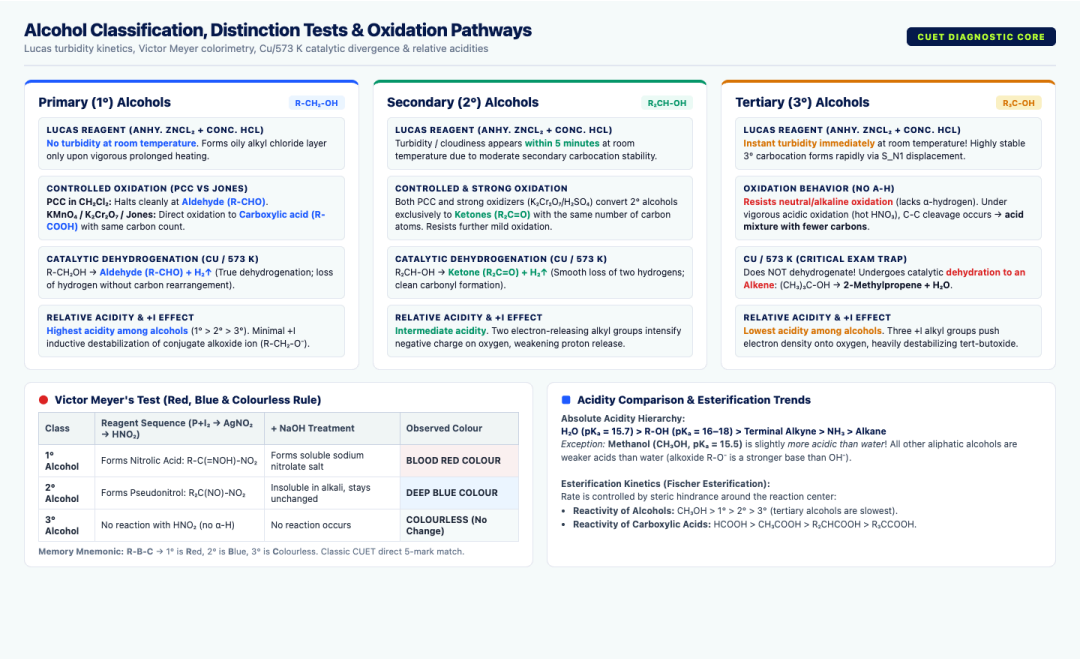


Figure 1: Alcohol classification, diagnostic distinction tests (Lucas turbidity kinetics & Victor Meyer colorimetry), Cu/573 K catalytic divergence, and relative acidities.

Module 3: Physical Properties & Distinction Tests of Alcohols

1. Physical Properties Trends

- **Boiling Points:** Alcohols have strikingly higher boiling points than alkanes, haloalkanes, or ethers of comparable molecular mass:

Ethanol (M = 46, b.p. = 351 K) versus Propane (M = 44, b.p. = 231 K) versus Methoxymethane (M = 46, b.p. = 248

Explanation: Intermolecular hydrogen bonding ($O-H\cdots O$) links alcohol molecules into extensive dynamic polymeric aggregates, requiring substantial thermal energy to vaporise.

- **Effect of Branching on Isomers:** Boiling point decreases with chain branching:

n-butyl alcohol (391 K) > isobutyl alcohol (381 K) > sec-butyl alcohol (373 K) > tert-butyl alcohol (356 K)

Branching diminishes surface area, making the molecule more compact and weakening van der Waals dispersion forces.

- **Water Solubility:** Lower alcohols (methanol, ethanol, propanol) are completely miscible with water in all proportions because their hydroxyl groups form strong hydrogen bonds with water molecules. Solubility drops sharply as the non-polar, hydrophobic hydrocarbon chain extends (C_5 and higher).

2. Chemical Properties & Acidity of Alcohols

Alcohols act as versatile amphoteric compounds (both weak Brønsted acids and Lewis bases):

1. **Acidity Hierarchy:**

$$\mathrm{H_2\ O}\ (pK_a = 15.7) > R-\mathrm{OH}\ (pK_a = 16-18) > \text{Terminal Alkynes} > \mathrm{NH_3} > \text{Alkanes}$$

Crucial Exception: Methanol ($\mathbf{CH_3\ OH}$, $\boldsymbol{pK_a = 15.5}$) is slightly **more acidic than water**! All other aliphatic alcohols are weaker acids than water. Consequently, alkoxide ions ($R-O^-$) are stronger bases than hydroxide ion ($\mathrm{OH^-}$).

2. **Acidity Order Among Alcohol Classes:**

$$1^{\circ} \text{ Alcohol} > 2^{\circ} \text{ Alcohol} > 3^{\circ} \text{ Alcohol}$$

Mechanism: Alkyl groups are electron-releasing ($+I$ inductive effect). In tertiary alcohols, three alkyl groups pump electron density onto oxygen, intensifying the negative charge on the conjugate alkoxide ion and heavily destabilizing it.

3. **Esterification (Fischer Esterification):** Reaction of alcohols with carboxylic acids in the presence of concentrated H_2SO_4:

$$R - \text{COOH} + R' - \text{OH} \overset{\text{H}^+}{\leftrightarrow} R - \text{COOR'} + \text{H}_2\text{O}$$

- **Steric Factor Dictates Rate:** Reactivity of alcohols follows: CH_3 OH $> 1^{\circ} > 2^{\circ} > 3^{\circ}$. Bulkier tertiary alcohols esterify sluggishly.
- **Mechanism Fact:** The $C - O$ bond of the acid breaks, and the $O - H$ bond of the alcohol breaks (proven by ^{18}O isotopic labelling).

3. Diagnostic Chemical Distinction Tests

1. **The Lucas Test (Distinction via $C - O$ Cleavage Kinetics):** Uses Lucas Reagent (concentrated HCl and anhydrous $ZnCl_2$). Alcohols convert into insoluble alkyl chlorides, creating turbidity/cloudiness:
   - **Tertiary (3°) Alcohols:** Produce instant turbidity immediately at room temperature (forms stable 3° carbocation via S_N1).
   - **Secondary (2°) Alcohols:** Turbidity appears within 5 minutes at room temperature.
   - **Primary (1°) Alcohols:** Remain clear at room temperature; turbidity appears only upon prolonged heating.
2. **Victor Meyer's Test (Distinction via Colorimetry):** Sequence: Alcohol $+\frac{P}{I_2} \rightarrow$ Iodoalkane $+AgNO_2 \rightarrow$ Nitroalkane $+HNO_2 \rightarrow$ Alkali (NaOH):
   - **Primary (1°) Alcohol:** Forms nitrolic acid, which dissolves in NaOH to yield an intense **Blood Red colour**.
   - **Secondary (2°) Alcohol:** Forms pseudonitrol, which is insoluble in alkali and produces a **Deep Blue colour**.
   - **Tertiary (3°) Alcohol:** Lacks α-hydrogen; does not react with nitrous acid, remaining completely **Colourless**.

   CUET Memory Formula: R-B-C ($1^{\circ} = \text{Red}, 2^{\circ} = \text{Blue}, 3^{\circ} = \text{Colourless}$).
3. **Catalytic Dehydrogenation over Heated Copper ($\frac{\text{Cu}}{573}$ K):**
   - **Primary (1°) Alcohol:** Dehydrogenated cleanly to an **Aldehyde**:

$$R - \text{CH}_2\text{OH} \overset{\frac{\text{Cu}}{573}\text{ K}}{\rightarrow} R - \text{CHO} + \text{H}_2 \uparrow$$

   - **Secondary (2°) Alcohol:** Dehydrogenated to a **Ketone**:

$$R_2\text{CH-OH} \overset{\frac{\text{Cu}}{573}\text{ K}}{\rightarrow} R_2\text{C=O} + \text{H}_2 \uparrow$$

   - **Tertiary (3°) Alcohol (HIGH-YIELD TRAP):** Because it lacks an α-hydrogen atom, it **cannot dehydrogenate**. Instead, it undergoes catalytic **dehydration to an Alkene**:

$$(\text{CH}_3)_3C - \text{OH} \overset{\frac{\text{Cu}}{573}\text{ K}}{\rightarrow} (\text{CH}_3)_2C = \text{CH}_2 + \text{H}_2\text{O} \quad \text{(2-methylpropene)}$$

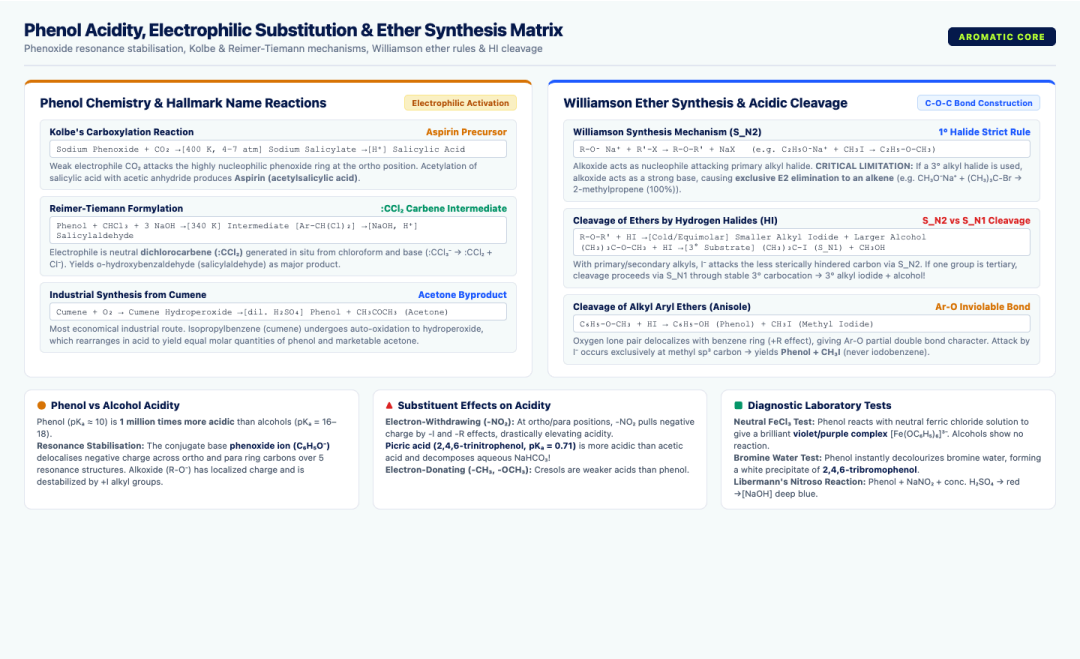


Figure 2: Phenols acidity mechanisms, phenoxide resonance stabilization, hallmark name reactions (Kolbe, Reimer-Tiemann), Williamson ether synthesis constraints, and HI cleavage regiochemistry.

Module 4: Chemical Properties & Name Reactions of Phenols

1. Acidity of Phenols & Substituent Effects

Phenol ($pK_a \approx 10$) is an acidic organic compound—approximately **one million times more acidic than aliphatic alcohols** ($pK_a \approx 16-18$):

- **Phenoxide Ion Resonance Stabilisation:** Ionisation yields a phenoxide ion ($C_6H_5O^-$). The negative charge is delocalised over the ortho and para positions of the aromatic ring across five contributing resonance structures. In contrast, the alkoxide ion ($R-O^-$) has its negative charge localized entirely on oxygen and is destabilized by $+I$ alkyl groups.
- **Phenol vs Carboxylic Acid:** Although phenol turns blue litmus red and dissolves in aqueous NaOH, it is a weaker acid than carbonic acid (H_2CO_3) and carboxylic acids ($R-COOH$). Hence, **phenol does NOT react with sodium bicarbonate ($NaHCO_3$)** and produces no effervescence of CO_2.
- **Substituent Influence on Phenol Acidity:**
  - **Electron-Withdrawing Groups ($-NO_2, -CN$, Halogens):** Disperse negative charge from the phenoxide ring via $-I$ and $-R$ effects, dramatically increasing acidity. The activating effect is maximal at **ortho** and **para** positions:

d) ($pK_a = 0.71$) > 2,4-dinitrophenol ($pK_a = 4.0$) > p-nitrophenol ($pK_a = 7.15$) > o-nitrophenol ($pK_a = 7.23$) > m-nitrophe

Examination Fact: Picric acid is so strongly acidic that it decomposes aqueous $NaHCO_3$, releasing effervescence of CO_2!

  - **Electron-Donating Groups ($-CH_3, -OCH_3, -NH_2$):** Intensify negative charge on the phenoxide ring via $+I$ and hyperconjugation/resonance, decreasing acidity:

Phenol ($pK_a = 10.0$) > p-cresol ($pK_a = 10.26$) > m-cresol ($pK_a = 10.08$) > o-cresol ($pK_a = 10.28$)

2. Electrophilic Aromatic Substitution of Phenols

The hydroxyl group is powerfully activating and strictly **ortho/para-directing**:

1. **Halogenation in Different Solvents:**
   - **In Non-Polar Solvent (CS_2 or $CHCl_3$ at 273 K):** Phenol is minimally ionised; monobromination yields **p-bromophenol (major)** and o-bromophenol (minor):

$$C_6H_5\ OH + Br_2 \xrightarrow{CS_2, 273\ K} \text{p-bromophenol (80\%)} + \text{o-bromophenol (20\%)}$$

   - **In Aqueous Medium ($\frac{Br_2}{H_2O}$):** Water ionises phenol completely into phenoxide ion ($C_6H_5O^-$), which hyperactivates the ring towards electrophilic attack. Instantaneous tribromination occurs, yielding a characteristic **white precipitate of 2,4,6-tribromophenol**:

$$C_6H_5\ OH + 3Br_2(aq) \rightarrow \text{2,4,6-tribromophenol} \downarrow \quad \text{(White Precipitate)} + 3\ HBr$$

2. **Nitration of Phenol:**
   - **With Dilute Nitric Acid (dil. HNO_3 at 298 K):** Yields a mixture of ortho-nitrophenol and para-nitrophenol:
     - **o-Nitrophenol:** Features **intramolecular hydrogen bonding** (chelation within the same molecule). It has a lower boiling point and is **steam volatile**.
     - **p-Nitrophenol:** Features **intermolecular hydrogen bonding** (linking separate molecules into aggregates). It has a higher boiling point and is **non-steam volatile**.

     Industrial Separation: The two isomers are cleanly separated by **steam distillation**.
   - **With Concentrated Nitric Acid ($\frac{\text{conc. } HNO_3}{\text{conc. } H_2SO_4}$):** Direct nitration produces **2,4,6-trinitrophenol (Picric acid)** in poor yields due to destructive oxidation of the ring.

3. Hallmark Name Reactions of Phenols

1. **Kolbe's Reaction (Synthesis of Salicylic Acid):** Phenoxide ion is treated with carbon dioxide (CO_2) at 400 K under $4-7$ atm pressure, followed by acidification:

$$C_6H_5\ ONa + CO_2 \xrightarrow{400\ K, 4-7\ atm} \text{o-hydroxybenzoic acid (Salicylic acid)}$$

   Pharmaceutical Gateway: Acetylation of salicylic acid with acetic anhydride in the presence of acid yields **Aspirin (acetylsalicylic acid)**:

$$\text{Salicylic Acid} + (CH_3\ CO)_2\ O \xrightarrow{H^+} \text{Acetylsalicylic acid (Aspirin)} + CH_3\ COOH$$

2. **Reimer-Tiemann Reaction (Synthesis of Salicylaldehyde):** Heating phenol with chloroform ($CHCl_3$) in aqueous sodium hydroxide at 340 K followed by acidic hydrolysis:

$$C_6H_5\ OH + CHCl_3 + 3\ NaOH \xrightarrow{340\ K} \text{Salicylaldehyde (o-hydroxybenzaldehyde)} + 3\ NaCl + 2H_2O$$

   Crucial Mechanistic Intermediates: The reactive electrophile is neutral **dichlorocarbene (: CCl_2)** generated in situ via α-elimination from chloroform:

$$CHCl_3 + OH^- \leftrightarrow :CCl_3^- + H_2O \rightarrow :CCl_2 + Cl^-$$

   If carbon tetrachloride (CCl_4) is used instead of chloroform, the reaction yields **Salicylic acid**.
3. **Reaction with Zinc Dust:** Distillation of phenol with zinc dust strips the oxygen, reducing it to **benzene**:

$$C_6H_5\ OH + Zn \xrightarrow{\Delta} C_6H_5 - H + ZnO$$

4. **Oxidation with Chromic Acid ($Na_2Cr_2\frac{O_7}{H_2SO_4}$):** Conjugated oxidation generates **p-benzoquinone**, a bright yellow conjugated diketone.

5. **Neutral Ferric Chloride Test ($FeCl_3$):** Phenols react with neutral $FeCl_3$ to form intensely coloured violet/purple coordination complexes: $\left[Fe(OC_6H_5)_6\right]^{3-}$. Aliphatic alcohols give no colour.

Module 5: Ethers — Synthesis, Properties & Acidic Cleavage

1. Methods of Preparation of Ethers

1. **Dehydration of Alcohols (Temperature-Controlled Divergence):** When ethanol is heated with concentrated sulphuric acid, the reaction outcome is controlled by temperature:
   - **At 413 K (140° C):** Intermolecular dehydration via bimolecular nucleophilic substitution (S_N2) yields **Diethyl ether**:
   $$2C_2H_5\ OH \xrightarrow{\text{conc. } H_2SO_4, 413\ K} C_2H_5 - O - C_2H_5 + H_2\ O$$
   - **At 443 K (170° C):** Intramolecular dehydration via β-elimination dominates, yielding **Ethene**:
   $$C_2H_5\ OH \xrightarrow{\text{conc. } H_2SO_4, 443\ K} CH_2 = CH_2 + H_2\ O$$
2. **Williamson Ether Synthesis (Universal Laboratory Route):** An alkyl halide is heated with a sodium alkoxide or phenoxide in an S_N2 displacement:
   $$R - O^-Na^+ + R' - X \rightarrow R - O - R' + NaX$$

   THE SACRED WILLIAMSON LIMITATION (MUST MASTER FOR CUET):
   - The alkyl halide $R' - X$ **must strictly be primary (1°)**!
   - If a **tertiary (3°) alkyl halide** is used, the alkoxide behaves as a strong base rather than a nucleophile, causing **exclusive E2 dehydrohalogenation to an Alkene**:
   $$CH_3O^-Na^+ + (CH_3)_3C - Br \rightarrow (CH_3)_2C = CH_2 (100\%\ \text{alkene}) + CH_3\ OH + NaBr$$
   - To prepare tert-butyl methyl ether, one must react **sodium tert-butoxide with methyl bromide**:
   $$(CH_3)_3C - O^-Na^+ + CH_3\ Br \rightarrow (CH_3)_3C - O - CH_3 + NaBr \quad \text{(Smooth S_N2 synthesis)}$$

2. Acidic Cleavage of Ethers by Halogen Acids ($\frac{HI}{HBr}$)

Ethers are relatively unreactive compounds, but their etherial $C - O$ linkage is cleaved under vigorous conditions by concentrated halogen acids (HI > HBr ≫ HCl):

1. **Cleavage of Symmetrical Ethers:**
   $$R - O - R + HI \rightarrow R - I + R - OH \xrightarrow{\text{excess HI}, \Delta} 2R - I + H_2\ O$$
2. **Cleavage of Unsymmetrical Ethers Containing Primary / Secondary Alkyl Groups (S_N2):** When an unsymmetrical ether containing 1° or 2° alkyl groups is cleaved with equimolar cold HI, the halide ion (I^-) attacks the **less sterically hindered carbon atom**:
   $$CH_3 - O - CH_2CH_2CH_3 + HI \rightarrow CH_3 - I + CH_3CH_2CH_2\ OH$$
3. **Cleavage of Ethers Containing a Tertiary Alkyl Group (S_N1):** If one of the alkyl groups is tertiary (3°), cleavage proceeds via an S_N1 mechanism through a highly stable tertiary carbocation. The halide ion attacks the tertiary carbon:
   $$(CH_3)_3C - O - CH_3 + HI \rightarrow (CH_3)_3C - I + CH_3\ OH$$

4. **Cleavage of Alkyl Aryl Ethers (Anisole):** When anisole ($C_6H_5 - O - CH_3$) is heated with concentrated HI, the products are exclusively **Phenol and Methyl Iodide**:

$$C_6H_5 - O - CH_3 + HI \rightarrow C_6H_5\ OH + CH_3I$$

Mechanism Justification: Oxygen lone pair resonance delocalises into the aromatic ring, giving the $Ar - O$ bond partial double-bond character. The $Ar - O$ bond is extraordinarily strong and cannot be cleaved; nucleophilic attack occurs strictly at the sp^3 methyl carbon. Iodobenzene is never formed.

Module 6: Commercially Important Industrial Alcohols

1. **Methanol (Methyl Alcohol / Wood Spirit, $CH_3\ OH$):**
   - Historically obtained by the destructive distillation of wood.
   - **Modern Industrial Synthesis:** Catalytic reduction of carbon monoxide (water gas) with hydrogen:

$$CO + 2H_2 \xrightarrow{ZnO - Cr_2O_3, 573-673\ K, 200-300\ atm} CH_3\ OH$$

   - **Physical Properties & Uses:** Colourless liquid ($b.p. = 337$ K). Used as an industrial solvent for paints and varnishes, and in the manufacture of formaldehyde (HCHO).
   - **Severe Toxicology:** Ingestion of even 10 mL causes permanent retinal blindness; 30 mL causes death. In the body, alcohol dehydrogenase oxidizes methanol to formaldehyde (HCHO) and formic acid (HCOOH), which coagulates cellular protoplasm and induces acute metabolic acidosis.
2. **Ethanol (Ethyl Alcohol / Grain Alcohol, $C_2H_5\ OH$):**
   - **Manufacture by Fermentation:** Prepared from sugary molasses or starchy grains using yeast enzymes:
     - Invertase converts sucrose into glucose and fructose:

$$C_{12}H_{22}O_{11} + H_2\ O \xrightarrow{Invertase} C_6H_{12}O_6 + C_6H_{12}O_6$$

     - Zymase ferments glucose/fructose into ethanol and carbon dioxide:

$$C_6H_{12}O_6 \xrightarrow{Zymase} 2C_2H_5\ OH + 2CO_2 \uparrow$$

     Fermentation terminates naturally when ethanol concentration reaches 14%, because higher concentrations poison the yeast enzyme.
   - **Denatured Alcohol (Methylated Spirit):** Commercial ethanol is rendered unfit for drinking by adding 5% toxic methanol, copper sulphate (imparting a warning blue colour), and pyridine (a foul-smelling liquid) to curb industrial tax evasion.

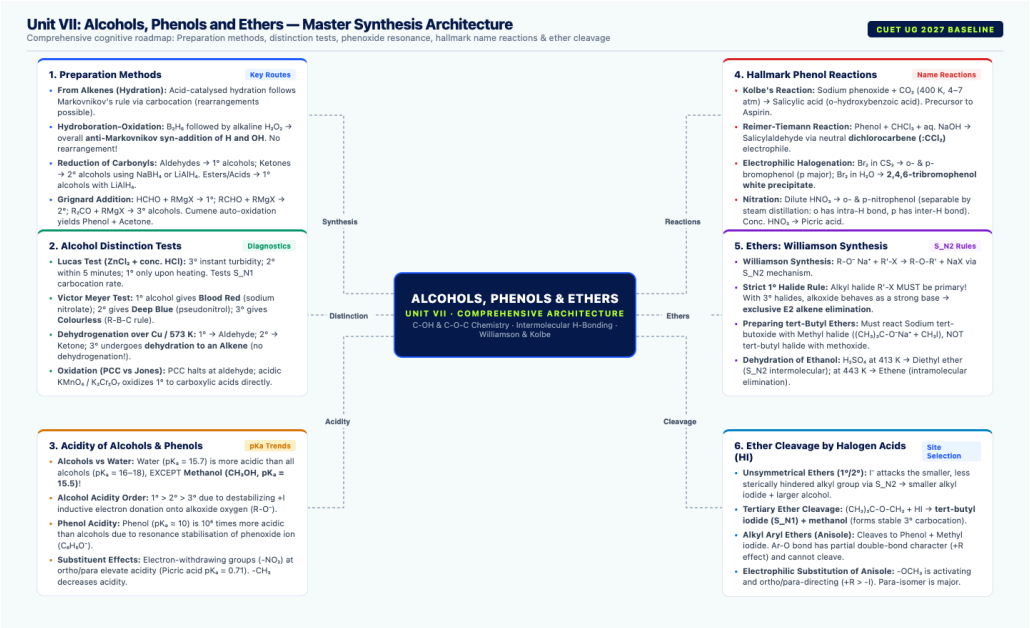


Figure 3: Master Synthesis Architecture: Complete Unit VII cognitive roadmap integrating alcohol distinctions, phenoxide resonance, hallmark name reactions, Williamson ether rules, and HI cleavage mechanisms.

UNIT VII MASTER FORMULA & HIGH-YIELD CHEATSHEET

Core Concept / Law	Expression / Formulation	Application & CUET Problem Strategy
Ether Bond Angle	$C-O-C = 111.7^\circ > 109.5^\circ$ (Tetrahedral)	Steric repulsion between two bulky alkyl groups widens the bond angle.
Hydroboration-Oxidation	Alkene $+B_2H_6 \xrightarrow{H_2O_2/OH^-}$ Alcohol	Overall anti-Markovnikov syn addition of water. Zero rearrangements.
Lucas Test Distinction	Conc. HCl + anhy. $ZnCl_2$ turbidity	3°: Instant turbidity; 2°: 5 minutes; 1°: Turbidity only on heating.
Victor Meyer (R-B-C)	Sequence: $\frac{P}{I_2} \rightarrow AgNO_2 \rightarrow HNO_2 \rightarrow NaOH$	1° = Blood Red; 2° = Deep Blue; 3° = Colourless (no reaction).
Dehydrogenation ($\frac{Cu}{573}$ K)	$1^\circ \rightarrow$ Aldehyde; $2^\circ \rightarrow$ Ketone; $3^\circ \rightarrow$ Alkene	Tertiary alcohol dehydrates to alkene; does not dehydrogenate!
Phenol Acidity	Phenol ($pK_a \approx 10$) is 10^6 times more acidic than alcohols	Phenoxide ion is resonance-stabilized. Picric acid ($pK_a = 0.71$) releases CO_2 from $NaHCO_3$.
Kolbe's Reaction	Sodium phenoxide $+CO_2 \xrightarrow{400\text{ K},4-7\text{ atm}} \xrightarrow{H^+}$ Salicylic acid	Synthesis of o-hydroxybenzoic acid; precursor to Aspirin.
Reimer-Tiemann Reaction	Phenol $+CHCl_3 + NaOH \xrightarrow{H^+}$ Salicylaldehyde	Electrophile is neutral dichlorocarbene ($:CCl_2$) from chloroform.
Phenol Tests	Neutral $FeCl_3 \rightarrow$ Violet complex; $\frac{Br_2}{H_2O} \rightarrow$ White ppt	Neutral $FeCl_3$ test $\left[Fe(OC_6H_5)_6\right]^{3-}$; 2,4,6-tribromophenol white precipitate.
Steam Distillation	o-nitrophenol (steam volatile) vs p-nitrophenol (non-volatile)	Ortho has intramolecular H-bonding; Para has intermolecular H-bonding.
Williamson Synthesis	$R-O^-Na^+ + R'-X \rightarrow R-O-R' +$ NaX	$R'-X$ must be 1°! Reaction with 3° halide gives 100% alkene elimination.
HI Ether Cleavage	Simple: I^- attacks less hindered carbon; 3°: forms 3° iodide	Anisole ($C_6H_5-O-CH_3$) gives Phenol $+CH_3I$ (Ar-O never cleaves).

Practice Questions (Exam Pattern) — Part 1 (Q1 to Q5)

Q1. An unknown alcohol reacts with Lucas reagent (concentrated HCl and anhydrous $ZnCl_2$) to produce instantaneous cloudiness / turbidity at room temperature. The alcohol is most likely:

(A) Butan-1-ol
(B) Butan-2-ol
(C) 2-Methylpropan-2-ol (tert-butyl alcohol)
(D) 2-Methylpropan-1-ol (isobutyl alcohol)

Sol. Q1. Lucas reagent reacts via an S_N1 mechanism wherein the rate is governed by carbocation stability. Tertiary alcohols form exceptionally stable 3° carbocations, undergoing immediate conversion into insoluble alkyl chlorides and producing instant turbidity at room temperature: 3° (instant) $> 2^\circ$ (5 mins) $> 1^\circ$ (heat only). 2-Methylpropan-2-ol is a 3° alcohol. **Answer: (C)**

Q2. In the Victor Meyer test for distinguishing alcohols, the colours produced by primary, secondary, and tertiary alcohols after treatment with alkali (NaOH) are respectively:

(A) Blue, Red, and Colourless
(B) Blood Red, Deep Blue, and Colourless
(C) Colourless, Blood Red, and Deep Blue
(D) Blood Red, Colourless, and Deep Blue

Sol. Q2. In the Victor Meyer test, primary alcohols form nitrolic acid which dissolves in alkali giving a characteristic Blood Red colour. Secondary alcohols form pseudonitrol which is alkali-insoluble and appears Deep Blue. Tertiary alcohols lack an α-hydrogen, failing to react with nitrous acid and remaining Colourless (**R-B-C Rule**). **Answer: (B)**

Q3. When vapours of 2-methylpropan-2-ol (tert-butyl alcohol) are passed over heated copper metal at 573 K (300° C), the organic product formed is:

(A) 2-Methylpropanal
(B) 2-Methylpropanoic acid
(C) 2-Methylpropene
(D) Butan-2-one

Sol. Q3. Primary and secondary alcohols undergo catalytic dehydrogenation over heated copper at 573 K to yield aldehydes and ketones respectively. However, tertiary alcohols lack an α-hydrogen atom on the hydroxyl-bearing carbon and cannot undergo dehydrogenation; instead, they undergo catalytic **dehydration** to eliminate water and form an alkene: $(CH_3)_3C-OH \xrightarrow[573]{Cu} K (CH_3)_2C = CH_2 + H_2\ O$. **Answer: (C)**

Q4. Propene is subjected to hydroboration-oxidation by treatment with diborane (B_2H_6) followed by alkaline hydrogen peroxide ($H_2 \frac{O_2}{OH^-}$). The major alcohol obtained is:

(A) Propan-2-ol
(B) Propan-1-ol
(C) Propane-1,2-diol
(D) A 50:50 mixture of propan-1-ol and propan-2-ol

Sol. Q4. Hydroboration-oxidation involves concerted syn-addition of borane (BH_3) across the double bond, followed by oxidative cleavage with alkaline H_2O_2. The net transformation corresponds to the stereospecific **anti-Markovnikov addition of water** without any carbocation formation or skeletal rearrangement, yielding pure propan-1-ol as the exclusive product. **Answer: (B)**

Q5. Which of the following compounds exhibits the HIGHEST acidic strength in aqueous solution?

(A) Phenol
(B) p-Nitrophenol
(C) p-Cresol
(D) 2,4,6-Trinitrophenol (Picric acid)

Sol. Q5. Picric acid contains three strongly electron-withdrawing nitro groups ($-NO_2$) at the two ortho and one para positions. Through powerful $-I$ and $-R$ effects, they delocalise and stabilize the conjugate phenoxide anion to an extraordinary extent ($pK_a = 0.71$). It is more acidic than acetic acid and easily decomposes aqueous sodium bicarbonate ($NaHCO_3$), releasing effervescence of CO_2.
Answer: (D)

Practice Questions (Exam Pattern) — Part 2 (Q6 to Q10)

Q6. The reactive intermediate electrophile responsible for formylation in the Reimer-Tiemann reaction of phenol with chloroform in aqueous sodium hydroxide is:

(A) Formyl cation (H-$C^+ = O$) **(B)** Dichlorocarbene (: CCl_2)
(C) Trichloromethyl carbanion (: CCl_3^-) **(D)** Chloromethyl radical (CH_2 Cl)

Sol. Q6. In the Reimer-Tiemann reaction, hydroxide ion abstracts the acidic proton from chloroform ($CHCl_3$), followed by loss of a chloride ion (α-elimination) to generate the neutral, electron-deficient **dichlorocarbene (: CCl_2)** intermediate with an unfilled sextet of electrons. This carbene acts as the active electrophile attacking the phenoxide ring. **Answer: (B)**

Q7. A mixture of ortho-nitrophenol and para-nitrophenol obtained from the nitration of phenol with dilute nitric acid can be cleanly separated by steam distillation because:

(A) o-Nitrophenol possesses intramolecular hydrogen bonding and is steam volatile **(B)** p-Nitrophenol possesses intramolecular hydrogen bonding and is steam volatile
(C) o-Nitrophenol has a higher boiling point due to extensive intermolecular association **(D)** p-Nitrophenol is completely insoluble in boiling water

Sol. Q7. o-Nitrophenol forms a stable 6-membered ring via **intramolecular hydrogen bonding** between its hydroxyl and adjacent nitro groups (chelation). This prevents association with neighboring molecules, resulting in lower boiling point and high steam volatility. In contrast, p-nitrophenol forms **intermolecular hydrogen bonds** linking molecules into polymeric chains, conferring a high boiling point and non-steam volatility. **Answer: (A)**

Q8. Salicylic acid is converted into the globally utilized analgesic and anti-inflammatory drug Aspirin (acetylsalicylic acid) by treatment with:

(A) Methanol in the presence of concentrated sulphuric acid **(B)** Acetic anhydride in the presence of an acid catalyst
(C) Phenol in the presence of phosphorus oxychloride **(D)** Sodium hydroxide followed by carbon dioxide at 400 K

Sol. Q8. Aspirin is manufactured by acetylating the phenolic hydroxyl group of salicylic acid using acetic anhydride in the presence of catalytic concentrated sulphuric acid or phosphoric acid: $\text{o-HO-C}_6\text{H}_4 - \text{COOH} + (\text{CH}_3\ \text{CO})_2\ \text{O} \xrightarrow{\text{H}^+} \text{o-CH}_3\text{COO-C}_6\text{H}_4 - \text{COOH} + \text{CH}_3\ \text{COOH}$.
Answer: (B)

Q9. In Williamson ether synthesis, treating sodium methoxide ($CH_3O^-Na^+$) with 2-bromo-2-methylpropane (tert-butyl bromide) results in the formation of:

(A) tert-Butyl methyl ether **(B)** 2-Methylpropene (Isobutylene)
(C) Dimethyl ether **(D)** Di-tert-butyl ether

Sol. Q9. Williamson ether synthesis is an S_N2 displacement. Alkoxide ions are both strong nucleophiles and powerful Brønsted bases. When confronted with a sterically hindered tertiary alkyl halide, the alkoxide behaves exclusively as a base, abstracting a β-hydrogen in an E2 elimination to yield **2-methylpropene (100%)** rather than the substitution ether product. **Answer: (B)**

Q10. When anisole (methoxybenzene, $C_6H_5 - O - CH_3$) is heated with an equimolar amount of concentrated hydroiodic acid (HI), the products formed are:

(A) Iodobenzene and Methanol **(B)** Phenol and Methyl Iodide
(C) Phenol and Methanol **(D)** Iodobenzene and Methyl Iodide

Sol. Q10. In anisole, the oxygen lone pair is delocalised into the aromatic ring ($+R$ resonance effect), imparting significant partial double-bond character to the $Ar - O$ bond (136 pm). The $Ar - O$ bond is exceptionally strong and cannot be cleaved. The iodide ion (I^-) attacks the weaker sp^3 methyl carbon via S_N2, yielding **Phenol and Methyl Iodide**. Iodobenzene is never formed.
Answer: (B)

Practice Questions (Exam Pattern) — Part 3 (Q11 to Q15)

Q11. Cleavage of 2-methoxy-2-methylpropane ($(CH_3)_3C-O-CH_3$) with one equivalent of concentrated hydroiodic acid (HI) produces:

(A) tert-Butyl alcohol and Methyl iodide **(B)** tert-Butyl iodide and Methanol
(C) 2-Methylpropene and Methanol **(D)** Equal molar amounts of iodomethane and iodoethane

Sol. Q11. Unlike primary and secondary ethers which cleave via S_N2, ethers containing a tertiary alkyl group undergo cleavage via an S_N1 mechanism. Protonation of the ether oxygen is followed by heterolytic cleavage to generate the extraordinarily stable tertiary carbocation ($(CH_3)_3C^+$). Attack by iodide ion on this carbocation yields **tert-butyl iodide ($(CH_3)_3C-I$)** and **methanol (CH_3 OH)**. **Answer: (B)**

Q12. The experimental $C-O-C$ bond angle in dimethyl ether (111.7°) is noticeably larger than the normal tetrahedral angle (109.5°). This widening is primarily caused by:

(A) Resonance between carbon and oxygen orbitals **(B)** Steric repulsion between the two bulky methyl groups
(C) Strong hydrogen bonding between adjacent ether molecules **(D)** Attraction between the lone pairs of oxygen and methyl hydrogens

Sol. Q12. Although the oxygen atom in ethers is sp^3-hybridised with two non-bonding lone pairs, the steric hindrance and electrostatic electron-cloud repulsion between the two bulky alkyl groups attached to oxygen push them apart, expanding the $C-O-C$ bond angle from the tetrahedral value of 109.5° to 111.7°. **Answer: (B)**

Q13. Phenol reacts with aqueous bromine water ($\frac{Br_2}{H_2}$ O) and with bromine in carbon disulphide ($\frac{Br_2}{CS_2}$) at 273 K to give respectively:

(A) p-Bromophenol and 2,4,6-tribromophenol **(B)** 2,4,6-Tribromophenol (white ppt) and p-bromophenol (major)
(C) o-Bromophenol and m-bromophenol **(D)** 2,4,6-Tribromophenol and bromobenzene

Sol. Q13. In aqueous medium, water facilitates extensive ionisation of phenol into phenoxide ion ($C_6H_5O^-$), which hyperactivates the benzene ring to undergo polybromination, precipitating 2,4,6-tribromophenol as a white solid. In non-polar CS_2, phenol does not ionise; its ring is less activated, undergoing monobromination to yield p-bromophenol as the major product. **Answer: (B)**

Q14. In the commercial manufacturing process of Phenol from Cumene (isopropylbenzene), the valuable organic byproduct obtained in equimolar yield is:

(A) Acetaldehyde **(B)** Benzaldehyde
(C) Acetone (Propan-2-one) **(D)** Formic acid

Sol. Q14. In the cumene process, cumene is air-oxidized to cumene hydroperoxide, which undergoes acid-catalysed rearrangement with dilute sulphuric acid to cleave into equimolar amounts of **Phenol** and **Acetone (propan-2-one)**: $C_6H_5-C(CH_3)_2-OOH \xrightarrow{H^+} C_6H_5\ OH + CH_3COCH_3$. The simultaneous recovery of acetone makes this the world's primary industrial process. **Answer: (C)**

Q15. Commercial ethanol is intentionally converted into 'Denatured Alcohol' (Methylated Spirit) before sale by adding small quantities of:

(A) Water and sugar **(B)** Copper sulphate, pyridine, and methanol
(C) Acetic acid and sodium chloride **(D)** Chloroform and diethyl ether

Sol. Q15. To prevent the illicit diversion and consumption of duty-free industrial alcohol, commercial ethanol is 'denatured' (rendered poisonous and foul-tasting) by blending it with toxic methanol (approx. 5%), pyridine (imparting an intolerable noxious odour), and copper sulphate (giving a warning blue colouration). **Answer: (B)**

© OurMocks · www.ourmocks.com · CUET UG Chemistry · Unit VII: Alcohols, Phenols and Ethers · 2027 Examination Baseline